

Inclusive AI Fashion Stylist with Measurement and Color-Blind Safety Analysis

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Project Proposal Report

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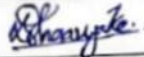
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Color plays an important role in daily activities such as selecting and matching clothes. However, individuals with Color deficiency often experience difficulty distinguishing certain colors, which can make choosing suitable clothing combinations challenging. Existing assistive applications mainly focus on detecting and naming colors through mobile cameras. Although these tools help users identify individual colors, they do not support decision-making when selecting matching outfits. Furthermore, most fashion recommendation systems are developed for people with normal vision and rarely consider the specific color perception limitations faced by color-blind users. Therefore, there is a clear gap in systems that assist color-blind individuals in evaluating clothing combinations in a practical and accessible way. To address this issue, this research proposes a color-blind friendly clothing pairing system that assists users in selecting suitable clothing combinations using mobile image analysis. The proposed system allows users to scan two clothing items using a camera. Image processing techniques are used to detect the dominant colors of the clothing items. A lightweight machine learning model will then be trained using a dataset of clothing images and labeled color combinations to classify whether the detected color pair is visually distinguishable for different color-blind types. Based on this analysis, the system provides simple feedback indicating whether the outfit combination is clear, acceptable, or potentially confusing. The system will be implemented using technologies such as OpenCV/YOLO for color extraction, Python for data processing and model training, and a mobile interface developed with React Native. The expected outcome is an assistive application that helps color-blind users confidently select clothing combinations without relying on external assistance. From a social perspective, the system supports accessibility and independent living, while commercially it has potential to be developed into a practical mobile tool for people with color vision deficiencies.

Keywords: Color deficiency, outfit matching, image processing, assistive technology

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
CV	Computer Vision
RGB	Red Green Blue Color Model
ML	Machine Learning
UI	User Interface
YOLO	You Only Look Once (Object Detection Model)
WBS	Work Breakdown Structure
SDG	Sustainable Development Goal
FR	Functional Requirement
NFR	Non-Functional Requirement

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Appendix	Description
Appendix A	Expert Consultation Letter
Appendix B	Color Deficiency Testing Tools Images (Ishihara Test Plates)

1. INTRODUCTION

1.1 Background and Context

Color plays an important role in everyday activities such as selecting and matching clothes. However, individuals with Color blindness often experience difficulty distinguishing certain colors, particularly combinations such as red–green or blue–yellow. Because of this limitation, color-blind individuals may unintentionally select clothing items that appear very similar or visually confusing. This issue can reduce confidence and independence when choosing outfits.

Color vision deficiency affects a considerable portion of the global population. Research shows that approximately 8% of men and around 0.5% of women worldwide are affected by some form of color blindness. Despite this, many existing clothing recommendation systems and fashion applications are designed mainly for people with normal color vision.

Several mobile applications have been developed to help color-blind users by detecting and naming colors using smartphone cameras. However, these systems usually focus on identifying a single color rather than helping users decide whether two clothing items form a suitable combination. As a result, color-blind users still face difficulties when selecting matching outfits.

To address this issue, this research proposes a mobile-based system that assists color-blind users in pairing clothing items using image processing and color analysis techniques. The system will detect the dominant colors of clothing items and evaluate whether the combination may appear distinguishable for different color-blind types. This study also aligns with the United Nations Sustainable Development Goal 10 by promoting accessibility and supporting individuals with visual perception limitations. The research falls within the area of assistive technologies and intelligent systems.

1.2 Literature Review

Several studies have investigated technological solutions to support individuals with color vision deficiency using image processing, computer vision, and assistive mobile applications. Color vision deficiency affects a significant portion of the global population and can influence daily activities that depend on color recognition, such as selecting clothing or identifying objects [1]. Research has also shown that approximately 8% of males and a smaller percentage of females worldwide are affected by red–green color vision deficiency, highlighting the importance of developing assistive technologies for this group [2].

A number of mobile applications have been developed to help color-blind users identify colors using smartphone cameras. These systems typically use image processing techniques to capture an image and analyze pixel values in order to determine the dominant color of an object. For example, Valdoria et al. developed an Android-based application that detects colors from captured images and provides color names to help color-blind individuals recognize objects more easily [3]. Similarly, Wong et al. proposed a computer vision approach for detecting shirt colors using image processing techniques, enabling users to identify clothing colors through automated analysis [4]. These solutions demonstrate the usefulness of camera-based color recognition technologies for assisting individuals with color vision deficiency.

In addition to simple color detection systems, several studies have explored fashion recommendation and clothing analysis technologies. Pottekat et al. discussed the development of outfit recommendation systems that evaluate clothing compatibility using color matching algorithms and fashion rules [5]. These systems typically analyze clothing attributes such as color, style, and patterns to recommend visually appealing combinations. Research by Li et al. also examined methods for evaluating color harmony in clothing combinations and generating recommendations based on aesthetic compatibility [8]. While these approaches help users select visually pleasing outfits, they are mainly designed for users with normal color perception and rarely consider the specific challenges faced by color-blind individuals.

Computer vision techniques have also been applied to clothing recognition and assistive technologies. Silva et al. introduced a system that uses computer vision and RFID technology to identify garments and assist visually impaired users in organizing clothing items [6]. Additionally, surveys of assistive technologies highlight the growing role of computer vision in supporting individuals with visual impairments through object detection, color recognition, and navigation assistance [7]. Reviews of computer vision applications in the fashion industry further demonstrate how image analysis can be used to recognize clothing attributes and support intelligent fashion systems [10].

From a technical perspective, many of these systems rely on image processing and machine learning techniques. Methods described in digital image processing research use pixel-level analysis and color space models to identify dominant colors within images [11]. Modern computer vision algorithms also enable object detection and feature extraction that support automated analysis of clothing images [12]. These techniques provide the technical foundation for developing intelligent clothing analysis systems.

Although previous research has made significant progress in color detection and fashion recommendation technologies, most existing systems focus either on identifying individual colors or generating fashion suggestions for general users. Only limited work has specifically addressed the problem of evaluating clothing combinations for individuals with color vision deficiency. Therefore, there is an opportunity to develop a system that integrates color detection with color-blind perception analysis to support clothing pairing decisions for users with color vision deficiency.

1.3 Research Gap

Although several technologies have been developed to support people with Color blindness, most existing solutions mainly focus on identifying individual colors rather than helping users decide whether two clothing items form a suitable outfit. Many mobile applications, such as Color Blind Pal, allow users to scan objects and receive the detected color name. While this feature helps users recognize colors, it does not assist them in determining whether two clothing colors are visually distinguishable when combined as an outfit.

Additionally, several research studies and fashion recommendation systems use computer vision and machine learning techniques to analyze clothing attributes and suggest outfits. However, these systems are generally designed for users with normal vision and do not consider the color perception differences associated with various types of color blindness, such as protanopia, deuteranopia, and tritanopia. As a result, color-blind users may still face

difficulties when selecting clothing combinations, even if the individual colors are correctly identified.

Furthermore, many existing solutions either require complex algorithms, additional hardware devices, or focus on general fashion matching rather than accessibility. This makes them less practical for a simple mobile-based assistive tool that specifically supports color-blind individuals in everyday clothing selection.

Therefore, there is a clear need for a lightweight and practical system that not only detects clothing colors but also evaluates whether two clothing items form a distinguishable combination for different color-blind conditions. The proposed research addresses this gap by introducing a color-blind friendly outfit pairing approach using color confusion mapping, which analyzes clothing color pairs and evaluates their suitability for specific color-blind types. This approach provides a simple yet innovative way to support independent clothing selection for color-blind users.

2. OBJECTIVE

2.1 Main Objective

To develop a system that detects the colors of two clothing items and evaluates whether the combination is distinguishable for different types of color-blind users

2.2 Specific Objectives

1. To develop an image processing module to detect the dominant color of clothing items using a mobile camera.
2. To classify detected clothing colors into basic color categories for easier analysis.
3. To identify the user's color-blind type and store the information in the system.
4. To design a color confusion mapping mechanism to analyze whether two clothing colors may appear similar for different color-blind conditions.
5. To implement a clothing pairing evaluation system that classifies outfit combinations as clear, acceptable, or low contrast for color-blind users.

3. METHODOLOGY

3.1 Research Approach

This study follows a Design Science research approach because the main goal is to design and develop a practical technological solution that assists individuals with Color blindness in selecting clothing combinations. The research focuses on creating a mobile-based system that detects clothing colors and evaluates whether two clothing items form a distinguishable outfit for different color-blind types. The approach involves designing the system, implementing the functions, and evaluating its effectiveness through testing.

3.2 Technical Methods and Algorithms

Step 1 – Image Capture

Users capture images of two clothing items (top and bottom) using the mobile application camera.

Step 2 – Clothing Color Detection

Image processing techniques are used to detect the dominant color of each clothing item. The system analyzes the pixel values of the image and identifies the most frequent color.

Step 3 – Color Categorization

Detected colors are mapped into basic color categories such as red, green, blue, yellow, black, white, and gray.

Step 4 – User Color-Blind Type Identification

The user selects their color-blind type (protanopia, deuteranopia, or tritanopia) during registration.

Step 5 – Color Confusion Mapping

A predefined color confusion table is used to identify color pairs that are commonly difficult for different color-blind types.

Step 6 – Outfit Pairing Evaluation

The system compares the two clothing colors with the confusion table and classifies the outfit combination as:

- Clear Combination
- Acceptable Combination
- Low Contrast Combination

3.3 Tools and Platforms with Justification

The system will be developed using the following technologies:

- OpenCV/YOLO – Used for image processing and color detection.
- Python – Used for implementing image analysis algorithms and color classification logic.
- React Native – Used to develop the mobile application interface.
- MySQL or Firebase – Used to store user information and color confusion mapping data.

These technologies are selected because they provide efficient image processing capabilities and support cross-platform mobile development.

3.4 Data Collection and Ethical Considerations

Clothing images will be collected by capturing photos of different colored clothing items during system testing. The dataset will include various clothing colors to ensure accurate color detection.

User data such as color-blind type will be collected only for system functionality. Personal information will be stored securely and used only for research purposes. Participants involved in testing will be informed about the purpose of the study, and their consent will be obtained before data collection.

3.5 Validation Strategy and Performance Metrics

The system will be evaluated through testing with multiple clothing combinations. The performance will be assessed using the following criteria:

- Color Detection Accuracy – Correct identification of clothing colors.
- Pairing Classification Accuracy – Correct classification of outfit combinations based on color confusion rules.

Testing will involve evaluating multiple clothing pairs and verifying whether the system correctly identifies combinations that may be confusing for color-blind users.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

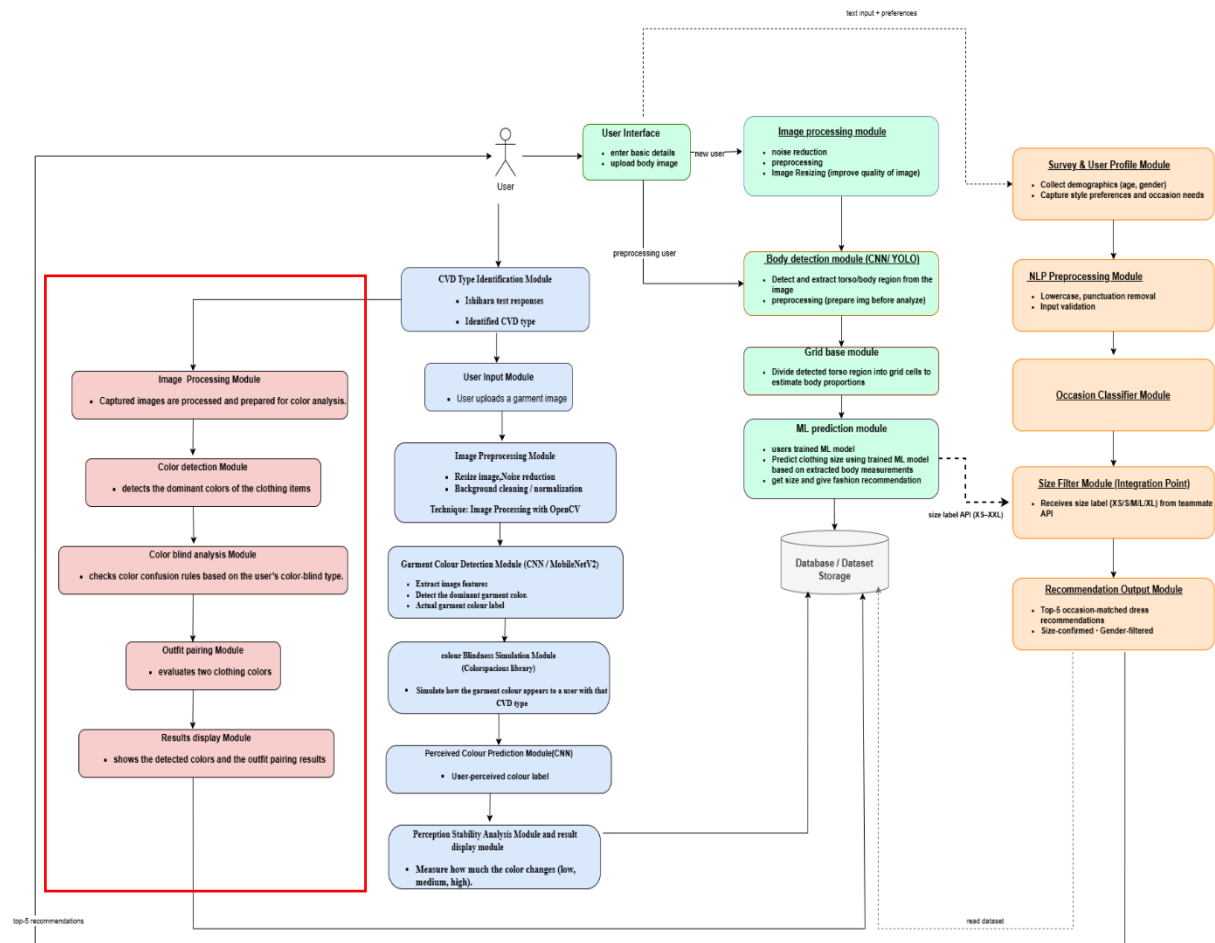


Figure 4.1: Overall System Diagram

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The primary stakeholders of the proposed system are individuals with Color blindness, who experience difficulty distinguishing certain colors when selecting clothing combinations. Secondary stakeholders include developers and researchers who design and maintain the application, as well as clothing retailers or fashion service providers who may benefit from integrating accessibility features into their platforms.

5.2 Functional Requirements

1. FR1: The system shall allow users to capture images of clothing items using the mobile device camera.
2. FR2: The system shall detect and identify the dominant color of the scanned clothing item.
3. FR3: The system shall allow users to scan two clothing items (top and bottom) for outfit pairing evaluation.
4. FR4: The system shall allow users to select or specify their color-blind type (e.g., protanopia, deuteranopia, or tritanopia).
5. FR5: The system shall analyze the detected clothing colors and compare them using a predefined color confusion mapping.
6. FR6: The system shall evaluate the outfit combination and classify it as clear, acceptable, or low contrast for the selected color-blind type.
7. FR7: The system shall display the detected clothing colors and the outfit pairing result to the user.

5.3 Non-Functional Requirements

1. Security-The system shall protect user data and prevent unauthorized access to stored information.
2. Performance-The system shall process clothing images and provide pairing results within a few seconds.
3. Usability-The mobile interface shall be simple and easy to use for individuals with visual perception limitations.
4. Reliability-The system shall provide consistent and accurate color detection results under normal lighting conditions.
5. Compatibility-The application shall operate on common smartphone platforms such as Android devices.
6. Accessibility-The interface design shall support clear text and simple visual feedback suitable for color-blind users.

6. COMMERCIALIZATION PLAN

The proposed application targets individuals with color blindness, as well as users who experience difficulty distinguishing clothing colors when selecting outfits. The typical customer persona includes users such as students and professionals who want a simple tool to help them match clothes independently. The value proposition of the system is to provide an accessible mobile solution that detects clothing colors and evaluates outfit combinations specifically for color-blind perception, which is not commonly available in existing color identification apps.

The application can adopt a freemium revenue model, where basic features are provided for free while advanced features such as personalized outfit suggestions are available through a small subscription. Development costs mainly involve software development, cloud storage, and maintenance, making it relatively low-cost compared to hardware-based assistive solutions. The pricing strategy would focus on affordability to encourage adoption among a wide user base.

In addition to the business-to-consumer (B2C) model, the system can also support a business-to-business (B2B) model by collaborating with clothing retailers, fashion platforms, and e-commerce companies. These businesses could integrate the color-blind outfit pairing feature into their shopping applications or websites to improve accessibility for customers with color vision deficiency. This integration could be offered as a licensing service or accessibility feature for partner platforms.

7. BUDGET AND JUSTIFICATION

Table 7.1: Budget

Item	Estimated Cost (LKR)	Reason
Cloud Services and Storage	Rs. 35,000	To store clothing images and support system deployment
Internet and Data Usage	Rs. 18,000	Required for development, testing, and cloud access
Dataset Collection and Preparation	Rs. 25,000	Collecting and organizing clothing image data
Testing Materials (Clothing Items)	Rs. 25,000	Different colored clothing items for testing outfit combinations
Cloud Computing for Model Training	Rs. 35,000	Required for training and testing the machine learning model
Travel for Data Collection / Discussion	Rs. 2,000	Small cost for visiting and discussing with a relevant person

Total Estimated Cost: Rs. 140,000

8. WORK BREAKDOWN STRUCTURE (WBS)

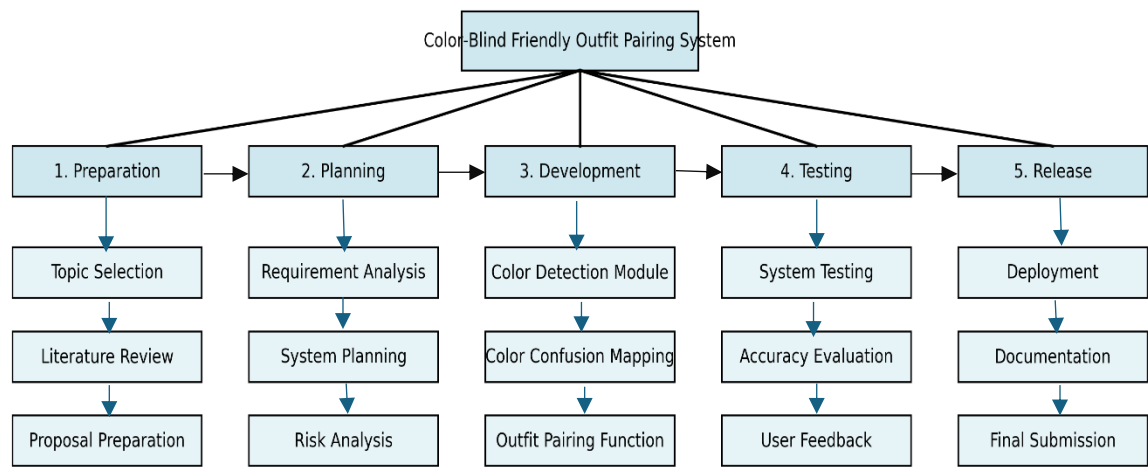
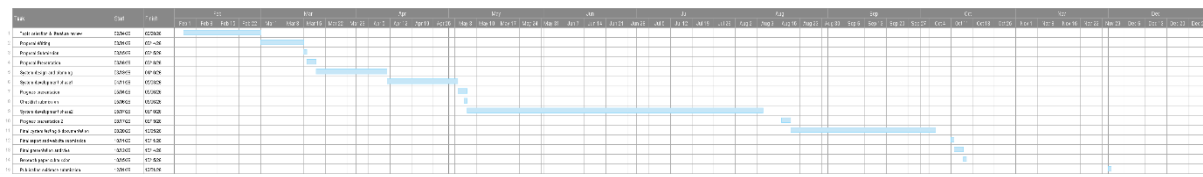


Figure 8.2: Work Breakdown Structure

9. GANTT CHART



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11.APPENDIX A. Expert Consultation Letter

This appendix includes a confirmation letter from an individual consulted during the initial stage of the research.

13/03/2026

Information Confirmation Letter.

I hereby confirm that I have provided information and shared my experiences regarding colour vision deficiency for the research study conducted by students of SLIIT. (IT22242372, IT22098696).

I understand that the information I shared will be used only for academic research purposes. I voluntarily participated in this discussion to support the study.

Name: Vidya Jayarathne, Head of optometry.

Signature: 

13/03/2026.

12.APPENDIX B. Color Deficiency Testing Tools Image

The following image was captured during the visit to illustrate the testing tools used to identify color deficiency conditions.

